

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 3: Application Based Questions**

**COURSE NAME:** Cloud Computing and Big Data Analytics      **COURSE CODE:** CSA15

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**COURSE OUTCOME COVERED**

**CO2:** *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2)  
Question Count: 5

**Total Marks: 25**

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**Code of Conduct**

I **JAIVANT.S (192311326)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **JAIVANT.S**

**Assessment Tasks**

**Application Based Questions**

1. How would virtualization help a company reduce hardware cost and improve utilization in a new cloud datacenter?
2. A startup needs rapid server provisioning for testing and deployment. Explain how hypervisors and virtual machines support this requirement.
3. How can SOA help integrate cloud services developed by different teams? Explain with an application-oriented example.
4. Explain how identity management and privacy controls should be applied in a federated cloud used by partner organizations.
5. A business wants to migrate from isolated physical servers to a software-defined datacenter. Explain the practical benefits they can expect.

**Application Based Questions Rubric**

| Criteria                 | Weightage | Level 4 - Exemplary                                                  | Level 3 - Proficient        | Level 2 - Developing   | Level 1 - Beginning     |
|--------------------------|-----------|----------------------------------------------------------------------|-----------------------------|------------------------|-------------------------|
| Relevance to Application | 25%       | Response directly addresses the given use case with strong relevance | Mostly relevant response    | Partially relevant     | Weak relevance          |
| Use of Cloud Concepts    | 25%       | Applies appropriate                                                  | Mostly accurate application | Basic application only | Incorrect concept usage |

|                   |     |                                       |                                          |                              |                      |
|-------------------|-----|---------------------------------------|------------------------------------------|------------------------------|----------------------|
|                   |     | concepts accurately                   |                                          |                              |                      |
| Practical Insight | 20% | Shows strong real-world understanding | Shows reasonable practical understanding | Limited practical insight    | No practical insight |
| Completeness      | 15% | Response covers all required aspects  | Covers most aspects                      | Covers only basic aspects    | Incomplete response  |
| Clarity           | 15% | Clear and concise answer              | Generally clear                          | Some confusion in expression | Poor clarity         |

## ANSWERS

### 1. Virtualization and hardware cost reduction

Virtualization allows one physical server to run multiple virtual machines (VMs) instead of using one physical server for each application.

- Reduces the number of physical servers required.
- Improves CPU, memory, and storage utilization.
- Reduces power, cooling, and maintenance costs.
- Makes server management easier.
- Allows resources to be allocated dynamically based on demand.

Example: Instead of buying 10 physical servers for 10 applications, a company can use 2–3 powerful servers and run multiple VMs on them.

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### 2. Hypervisors and VMs for rapid server provisioning

A hypervisor is software that creates and manages virtual machines on a physical server.

- A VM can be created from a preconfigured template in minutes.
- Multiple VMs can run independently on the same physical server.
- Developers can quickly create separate environments for testing and deployment.
- VMs can be cloned, deleted, restarted, or moved easily.
- Resources such as CPU, RAM, and storage can be assigned as needed.

Example: A startup can create a VM with Linux, install its application, test it, and create another identical VM for deployment without purchasing new hardware.

### **3. SOA for integrating cloud services**

Service-Oriented Architecture (SOA) divides an application into independent services that communicate through standard interfaces.

This helps different teams develop and maintain services independently while still allowing them to work together.

Application example: Consider an online shopping application:

- Team 1 develops the User Authentication Service.
- Team 2 develops the Payment Service.
- Team 3 develops the Inventory Service.
- Team 4 develops the Order Service.

These services communicate using APIs/web services. The Order Service can call the Payment and Inventory services without knowing their internal implementation.

Benefit: Different teams can use different technologies while their services remain interoperable.

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### **4. Identity management and privacy in a federated cloud**

In a federated cloud, multiple partner organizations share cloud resources. Proper identity and privacy controls are essential.

- Use federated identity management so users can access services using their organization's identity.
- Implement Single Sign-On (SSO) for convenient and controlled access.
- Apply Role-Based Access Control (RBAC) so users only access authorized resources.
- Use strong authentication such as multi-factor authentication (MFA).
- Encrypt sensitive data during storage and transmission.
- Maintain audit logs to track user activities.
- Apply data-sharing and privacy policies between partner organizations.
- Immediately revoke access when an employee leaves an organization.

Example: Employees from Company A can access a shared cloud application, but their permissions are limited to the resources assigned to Company A.

## 5. Benefits of a Software-Defined Datacenter

A Software-Defined Datacenter (SDDC) manages computing, storage, and networking through software rather than depending mainly on manually configured physical hardware.

Practical benefits include:

- Better resource utilization: Resources can be dynamically allocated.
- Faster provisioning: Servers and networks can be created quickly using software.
- Automation: Repetitive administrative tasks can be automated.
- Scalability: Resources can easily be increased or decreased according to demand.
- Lower costs: Less physical hardware and manual maintenance are required.
- Centralized management: Infrastructure can be controlled from a common software platform.
- Improved flexibility: Applications can be moved or deployed across available infrastructure more easily.

Example: Instead of manually configuring a new physical server whenever demand increases, the company can automatically create virtual servers and allocate resources through the SDDC platform.